

MATH FOR AI SAFETY: AN INVITATION FOR MATHEMATICIANS

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ABSTRACT. Powerful AI systems raise questions that look, at first, like engineering: What should the system optimize? What has it learned? Why does it generalize? How will it behave on unfamiliar inputs? Each has a mathematical core. I organize this invitation by mathematical field, so you can turn straight to your own: logic and game theory for cooperation and bounded reasoning; probability and causality for agency and world-models; algebra and representation theory for learned features; analysis and geometry for generalization and training dynamics. New mathematics is needed to help make AI systems more legible, steerable, and cooperative. Each section ends with a concrete open problem, chosen to be small enough to start and close enough to the safety question to matter.

1. INTRODUCTION

1.1. A question you already know how to ask. Every mathematician knows the question well: *where should the next hour of thought go?* Prove the lemma now, or assume it and see what it implies? Keep attacking the conjecture, or pivot to hunting a counterexample? Look up what is known, or re-derive it yourself? Each is a wager placed under uncertainty against a finite budget of attention. A mathematician is, among other things, a professional allocator of bounded compute.

The same problem arises in artificial intelligence: an agent with finite time, memory, and reasoning power must constantly decide what to compute, what to believe, whom to trust, and when to act. How should an agent assign something like probabilities to statements it has not yet had time to prove or refute? This is the problem of *logical induction* [19]. How long a proof must an agent be willing to search for before it can safely cooperate with a stranger? This is a resource-bounded form of Löb’s theorem [12]. The question that quietly organizes your research life also organizes this survey.

1.2. Three kinds of mathematician. I would like to distinguish three stances a mathematician can take toward the arrival of capable AI. The *artisanal* mathematician does mathematics by and for humans: each theorem, definition, and proof is bespoke, valued for its beauty and for the understanding it produces. Until very recently, all mathematics was artisanal. The *industrial* mathematician embraces AI as a collaborator and an accelerant: the unit of progress shifts from the human-readable paper toward shared, machine-checked libraries such as Lean’s

`mathlib` [30] and toward AI systems that take part in proof and discovery themselves [49, 36, 43, 16]. The *civic* mathematician turns the same craft toward a third end: helping humanity adapt to, and steer, the disruptions caused by our own technologies.

This survey is an invitation to try the third stance. The stances are not exclusive: one can prove a theorem for its intrinsic beauty, and also formalize the proof, and also ask what it implies for AI safety. But the civic stance invites a new perspective on an old question: of all the theorems we might prove with our bounded compute, which ones deserve the effort?

1.3. What is our profession’s purpose? Mathematics is one of the ways society builds its future. Our students become AI engineers, our theorems become tools in their hands, and our definitions become the language they use to describe what they are building.

Already we defer to systems we do not fully understand. Among the many risks from developing powerful AI systems, one of the most alarming is *gradual disempowerment* [25]. Preventing AI harms is in part a *mathematical* problem, and the purpose of this survey is to recruit mathematicians to work on it.

1.4. Where mathematicians plug in. At a high level, an AI system is produced in five stages, beginning with defining the desired behavior and ending with deploying the trained system, after which unanticipated failures feed back into refining the earlier stages (Figure 1).

To illustrate the five stages, consider the problem of building a tool for proving inequalities in a proof assistant:

- *Define*: given hypotheses H and a proposed inequality I , the system should either produce a formally checkable proof of $H \Rightarrow I$ or say “unknown.”
- *Measure*: score the fraction of benchmark statements proved within a fixed time and proof-length budget.
- *Train*: show the system many proof traces, say sums-of-squares certificates, AM-GM arguments, and induction templates.
- *Test*: hold out new inequalities from specified families.
- *Deploy*: let the tool suggest lemmas inside a mathematician’s formalization workflow.

After deployment, the failures come back: It returns a valid proof, but of a statement weaker than intended: a gap in the *definition* of the desired behavior. Its pass rate is high because the benchmark is full of short template problems, while the mathematician needs one nontrivial lemma: a gap in what we *measured*. It is brittle on boundary cases such as equality and zero denominators, which were rare in the proof traces: a gap in what it was *trained* on. And it passes the holdout set because the generator reused the same normal forms and substitutions: a gap in the *test*. Each backward arrow is its own piece of mathematics: formal specification, scoring rules, distribution design, and adversarial test construction.

The inequality example is deliberately mathematical, but the same basic pipeline applies across many areas, whether it is an image classifier for a medical application, a control system for a self-driving car, or a content recommendation system. Machine-learning research often concentrates on the *training* and *testing* stages, while engineers focus on *deployment*. The earlier stages rely on mathematicians, philosophers, economists, and others to supply the *defining* and *measuring* inputs.

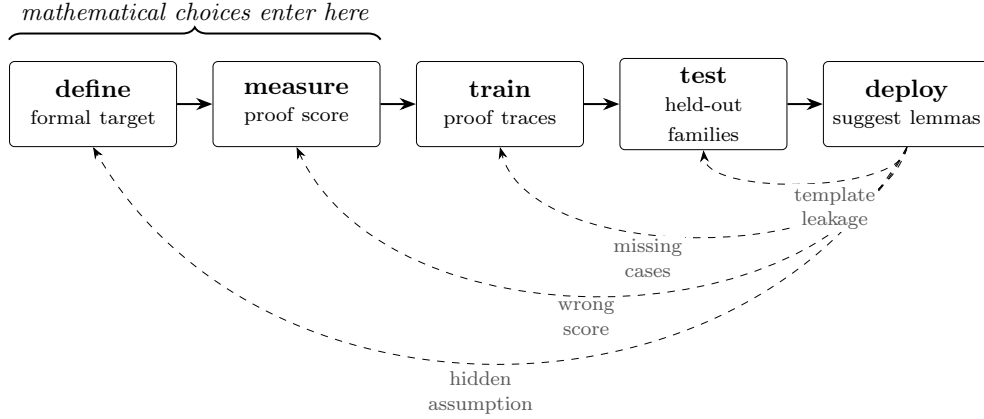


FIGURE 1. A schematic pipeline for producing an AI system, illustrated by a tool for proving inequalities. The forward arrows build the system; the dashed arrows represent deployment failures that inform improvements of each stage of the pipeline.

AI safety is sometimes posed as *control*—retaining power over systems more capable than ourselves—which looks like a very hard problem. *Coexistence*, a stable and mutually beneficial relationship with systems we do not fully control, is a larger target and thus an easier one to hit; much of the mathematics below is best read as aiming for coexistence rather than control.

This invitation is organized by mathematical field, so you can turn directly to yours. Each section states the safety stakes, narrates at least one theorem, and ends with at least one open problem (★). I keep machine-learning jargon to a minimum; a short glossary appears in Appendix A.

2. LOGIC AND GAME THEORY: WHEN DO AI AGENTS COOPERATE?

Here is a useful surprise: two programs that can read each other’s source code can be made to cooperate in a one-shot Prisoner’s Dilemma—provably, and (under explicit assumptions about the proof system) unexploitably, without checking that they are copies of one another. The tool is Löb’s theorem, a corner of mathematical logic rarely introduced through applications.

2.1. The game changes when the players can read each other. The Prisoner’s Dilemma is the standard model of a cooperation failure: each player does better by defecting whatever the other does, yet mutual cooperation (C, C) beats mutual

defection (D, D) . In the usual one-shot game, defection is dominant. But that analysis quietly assumes the players are *opaque* to each other. When two agents are programs, each can be handed the other’s source code before deciding. This is the setting of *open-source game theory*, or *program games*, studied since McAfee [31], Howard [22], and Tennenholtz [50].

One idea already breaks the classical verdict. Let CliqueBot be the program “cooperate if and only if the opponent’s source code is byte-for-byte identical to mine.” Two CliqueBots cooperate, and CliqueBot is never exploited, so (C, C) is reachable and stable. But CliqueBot is a poor citizen: it refuses to cooperate with an agent that is functionally identical yet formatted differently, or written in another language. We want cooperation that is *robust*—keyed to what the opponent *does*, not to the syntax of how it is written.

2.2. FairBot and the Löbian argument. Consider instead FairBot: *cooperate if and only if you can prove (in Peano Arithmetic, say) that the opponent cooperates with you*. Two FairBots reason about each other’s behavior, not each other’s syntax. Naively this looks like an infinite regress—each waits to prove something about the other, who is waiting in turn—and almost every mathematician, asked to guess, predicts mutual defection. The guess is wrong.

Theorem (Barász, Christiano, Fallenstein, Herreshoff, LaVictoire, Yudkowsky [4]). *FairBot cooperates with itself: $\text{PA} \vdash [\text{FairBot}(\text{FairBot}) = C]$. Moreover FairBot is unexploitable: assuming PA proves only true facts about the outputs of the programs in question, FairBot never cooperates with an opponent that defects against it.*

The proof is a simple application of Löb’s theorem, a strengthening of the self-reference behind Gödel’s second incompleteness theorem.¹ Write $\Box\varphi$ for the arithmetical sentence $\text{Prov}_{\text{PA}}(\ulcorner\varphi\urcorner)$, read as “ φ is provable in PA.”

Theorem (Löb [29]). *For any sentence φ , if $\text{PA} \vdash (\Box\varphi \rightarrow \varphi)$ then $\text{PA} \vdash \varphi$.*

Let φ be the sentence “both FairBots cooperate.” Each FairBot cooperates exactly when it finds a proof that the other does; so *if it is provable that both cooperate, then both genuinely do*—that is, $\text{PA} \vdash (\Box\varphi \rightarrow \varphi)$. Löb’s theorem now delivers $\text{PA} \vdash \varphi$ outright, and both cooperate!

FairBot as stated searches proofs of unbounded length. Critch [12] supplies a computable version, FairBot_k , that searches only proofs of length at most k (write $\Box_k\varphi$ for “ φ has a proof of length $\leq k$ ”), and shows it still cooperates with itself once k is large enough—turning the Löbian argument into a terminating computation and making the question quantitative: how large must k be?

2.3. Why a safety researcher cares—and worries. Read optimistically, this is a proof of concept that *sufficiently strong common knowledge can manufacture cooperation that classical rationality forbids*. Mutual transparency underwrites the binding commitments behind contracts, treaties, and trust; common knowledge is the raw material of cooperation [45].

¹In the special case $\varphi = \perp$, Löb’s theorem says that if PA proves its own consistency then PA proves a contradiction—Gödel’s second incompleteness theorem.

Read cautiously, the same mechanism is a hazard. Two AIs can be transparent to each other in ways no two humans can, so they can reach cooperative equilibria—including collusion *against* their human principals—that humans can neither match nor detect. And the map from transparency regime to outcome is full of surprises: Critch, Dennis, and Russell [13] exhibit natural institution designs whose large-resource behavior is the opposite of what one first expects. The design space is large, the failure modes are nonobvious, and nearly all of it is open.

Logical induction [19] is the probabilistic companion to this bounded-proof story. A *logical inductor* assigns prices $\mathbb{P}_n(\varphi)$ to arithmetical sentences on day n , while a slow deductive process reveals more theorems. The criterion is a bounded Dutch-book condition: no polynomial-time trader with bounded risk may make unbounded profit by buying and selling these sentence stocks. It is a theorem-level answer to the question this survey began with: how should a reasoner with finite computation allocate probability mass over facts it has not yet proved?

2.4. Open problems.

★ **Open Problem 1** (Quantitative bounded Löb). Fix a proof system S , a Gödel coding, and the length-bounded predicate $\Box_k^S$; write $S \vdash_{\leq k} Q$ for an S -proof of Q of length at most k . As a classical warm-up, quantify ordinary Löb: determine the least overhead F for which

$$S \vdash_{\leq k} (\Box_k^S P \rightarrow P) \quad \text{implies} \quad S \vdash_{\leq F(k, |P|)} P$$

holds uniformly, where $|P|$ denotes the length of the sentence. The bounded-agent version is the main target. In Critch’s theorem, a proof of $\forall t (\Box_{f(t)}^S p(t) \rightarrow p(t))$, with f large enough, yields a proof of $\forall t > \hat{t} (p(t))$. For natural program-game families $p(t)$, bound the smallest admissible budgets f , the threshold $\hat{t}$, and the resulting proof length in terms of the premise proof, the proof-system constants, and the agents’ code lengths. Ask also for lower bounds showing which overheads are unavoidable. For bounded FairBot, a clean first question is whether $\hat{t}$ is polynomial in the agent description length, in one standard proof system.

3. PROBABILITY AND CAUSALITY: WHAT IS AN AGENT?

The early stages of the pipeline—*define* and *measure*—need mathematical objects for words like belief, goal, and world-model. Probability supplies two especially useful pictures: an agent as something that maximizes expected utility, and an agent as something that keeps a probabilistic model of the hidden states behind its observations. Causality then asks when such a world-model can be recovered from behavior.

3.1. Two views of an agent. The first view is the *utility maximizer*. The agent has a current belief $q(z)$ over possible states z of the world and a utility function $u(\tau)$ assigning a number to each full outcome or trajectory τ . In a sequential setting, an “action” is a **policy** $\pi(a \mid h)$: a rule, possibly stochastic, for choosing an action a from the information history h available so far. Thus h is the past information the policy can condition on, while τ is the whole possible course of events being evaluated,

including the future. Together with the agent’s predictive beliefs, a policy induces a probability measure Q_π on trajectories, and the agent chooses π to maximize expected utility $\mathbb{E}_{\tau \sim Q_\pi}[u(\tau)]$. Von Neumann–Morgenstern [37] gives the classical representation theorem: if an agent’s preferences over lotteries satisfy a small list of consistency axioms—completeness, transitivity, continuity, and independence—then the agent behaves as if maximizing expected utility, with u unique up to positive affine transformation.

If several systems are competing for the same resources, and one of them leaves money on the table by having cyclic, incomplete, or otherwise exploitable preferences, then a system that makes the corresponding expected-utility tradeoffs will tend to beat it. Over time, this creates a selection pressure toward agents that are well modelled as maximizing some objective, even if they were not designed that way. The caveat is as important as the theorem. VNM assumes that the outcome space and the preferences are already given; it does not tell us what a trained network’s outcomes are, what its utility is, or whether its apparent preferences are complete and stable.

The second view is the *active predictor*. Here the central object is not a utility function but a probabilistic model of how observations are produced. Write z for the relevant state of the world and x for the present observation—for example, the pixels currently arriving at the retina. In this view z is something to be inferred from sensory data: objects, lighting, motion, and so on. A generative model is a joint density $p(x, z)$, read as “states z generate observations x .” This p is the agent’s standing world-model. The agent also maintains a distribution $q(z)$, its current belief about the hidden state, which it updates on receiving a new observation x . In predictive-coding models q is usually restricted to a simple family, and the agent tries to make q close to the exact posterior $p(z \mid x)$ implied by its world-model.

Here “close” is measured by *Kullback–Leibler divergence*. For probability densities r and s on the same space,

$$\text{KL}(r \parallel s) = \int r(z) \log \frac{r(z)}{s(z)} dz$$

This is the expected log-likelihood ratio in favor of r over s , when the data are actually drawn from r . Gibbs’ inequality says $\text{KL}(r \parallel s) \geq 0$, with equality if and only if $r = s$ almost everywhere. In information-theoretic terms, $\text{KL}(r \parallel s)$ is the mean excess surprise of an observer who sees a sample drawn from r but believes the sample was drawn from s . It is not symmetric, so it is not a metric; it is a directed measure of how badly s approximates r .

Predictive-coding models use this divergence through the *variational free energy*

$$F(q, x) = \mathbb{E}_q[\log q(z) - \log p(x, z)] = \text{KL}(q(z) \parallel p(z \mid x)) - \log p(x).$$

Since KL is nonnegative, this inequality says that $F(q, x)$ is an upper bound on the model’s surprise $-\log p(x)$ at seeing x . The surprise term does not depend on q . Thus, for fixed x , lowering $F(q, x)$ lowers this upper bound and makes q a better approximation to the posterior. In the special case where q is concentrated at one best guess and the noise is Gaussian, $F(q, x)$ becomes a sum of squared prediction

errors: the mismatch between what each layer of the model expected and what the layer below actually supplied. Updating the hidden variables to reduce these errors is a local error-correction rule. Rao and Ballard [40] proposed this as a model of visual cortex, and Millidge, Tschantz, and Buckley [32] show that such local predictive-coding updates can compute the same gradients as backpropagation—the chain-rule algorithm used to train neural networks—on an arbitrary directed graph.

	Agent as utility maximizer	Agent as active predictor
Core object	World-model $p(x, z)$ and utility $u(\tau)$	World-model $p(x, z)$ and variational family $\mathcal{Q}$
Belief update	Bayes: $q_{\text{new}}(z) \propto p(x z)q_{\text{old}}(z)$	Variational Bayes: $q_{\text{new}} \in \arg \min_{q \in \mathcal{Q}} F(q, x)$
Policy choice	$\pi^* \in \arg \max_{\pi} \mathbb{E}_{\tau \sim Q_{\pi}}[u(\tau)]$	$\pi^* \in \arg \min_{\pi} \mathbb{E}_{\tau \sim Q_{\pi}}[\sum_t F(q_t, x_t)]$
Notation	z : state x : observation $p(x, z)$: world-model $q(z)$: belief τ : trajectory π : policy	

TABLE 1. Two views of agency. The utility view starts with preferences, represents them by a utility function u , and asks which policy maximizes expected utility over the trajectories it may bring about. This is the natural language for proxy objectives and their failures—optimizing a measurable proxy until it diverges from the goal it stood for (Goodhart’s law, reward hacking). The active-predictor view starts with a world-model $p(x, z)$: the agent maintains an approximate belief q in an allowed family $\mathcal{Q}$, updates q by minimizing variational free energy, and chooses policies whose predicted futures have low expected free energy. Here apparent goals can live as priors over observations, so the safety question becomes: what world-model, priors, and inference procedure has the system learned?

Active inference is a candidate for a *descriptive* theory of agency: a way to recognize beliefs, goals, and decision-making structure in systems that were not hand-written as utility maximizers. In active inference, preferences can be represented as priors over observations—states of the world the agent expects, and acts to bring about. This fits trained neural networks naturally. A language model is already a probabilistic predictor; if some parts of its learned distribution function like persistent priors over outcomes, then goal-directed behavior may be implicit in the generative model rather than stored as a separate reward function. Ngo’s scale-free framing [35] pushes this further: the same analysis might apply to whole agents, to learned personas, or to subagent-like structures inside a network.

3.2. Robustness reveals causal structure. A *Bayesian network* is a directed acyclic graph G whose vertices are random variables $X_1, \dots, X_n$, together with

conditional probability tables satisfying

$$p(x_1, \dots, x_n) = \prod_i p(x_i \mid \text{pa}(x_i)),$$

where $\text{pa}(x_i)$ denotes the parents of X_i in G . The graph is a bookkeeping device for conditional independence: once the parents of a node are known, its non-descendants add no further information. Pearl’s graphical criterion, *d-separation*, reads off from the graph exactly which conditional independences are forced by this factorization [39].

A *causal* Bayesian network adds an interpretation to the arrows: each conditional distribution is a local mechanism. An intervention on a variable replaces its usual mechanism by a chosen value or distribution and leaves the other mechanisms alone. This is what distinguishes causation from correlation. If smoke and fire are correlated, observing smoke changes the probability of fire; intervening to make smoke with a smoke machine need not.

Richens and Everitt [42] make the world-model question precise using a *causal influence diagram*: a causal Bayesian network with extra decision nodes, where a policy chooses actions from observations, and utility nodes, whose values the agent is scored on. The agent is tested not just in one environment but across a rich family of local interventions, each of which resets some environment variable independently of its usual causes. Its regret under an intervention is the utility gap between its policy and the best policy for that intervened environment.

Theorem (Robust agents learn causal world models; Richens and Everitt [42]). *For almost all causal influence diagrams satisfying their regularity assumptions, any policy with regret at most δ across all local interventions determines an approximate causal model of the utility-relevant environment, with error bounded by a function $\gamma(\delta)$ satisfying $\gamma(0) = 0$ and growing linearly for small δ . In the optimal limit $\delta = 0$, the graph and joint distribution over all ancestors of the utility are identified exactly.*

The proof is constructive. Treat the policy as an oracle, mix two interventions by a parameter λ , and watch where the optimal action switches. The switch point is an indifference equation; because the utility is known, it solves for an interventional probability. Repeating this over local interventions recovers the conditional distributions and parent sets, except on measure-zero degeneracies such as exact cancellations or ties.

★ **Open Problem 2** (Behavioral tomography of world-models). Give finite-sample identifiability theorems for recovering an agent’s causal model from interventions on its inputs, *activations*, or environment. A clean first case is a finite causal influence diagram with binary variables and known utility, where one can query a policy under only a restricted set of interventions. Characterize which parts of the graph and conditional probability tables are identifiable, and how the error scales with regret, sample size, and the choice of interventions.

4. ALGEBRA: WHAT IS A LEARNED FEATURE?

Mechanistic interpretability tries to reverse-engineer a trained network into something a human can read. Much of its core is linear algebra and representation theory: how a network stores more concepts than it has dimensions, and what it means for a concept to be a *direction* one can read or steer.

4.1. Superposition: packing features into directions. How does a network with a few hundred dimensions in each layer represent the many thousands of distinct concepts (“features”) it seems to know? A leading hypothesis, made vivid by Elhage et al. [18], is *superposition*: when features are sparse, a network can store more of them than it has dimensions, as directions in activation space that are *not* orthogonal but only nearly so, tolerating the resulting interference because at any moment only a few features are active (Figure 2). In a small ReLU model one can watch the geometry organize itself into regular polytopes—antipodal pairs, triangles, pentagons—as a function of how sparse and how important the features are; it is, recognizably, a Thomson problem.

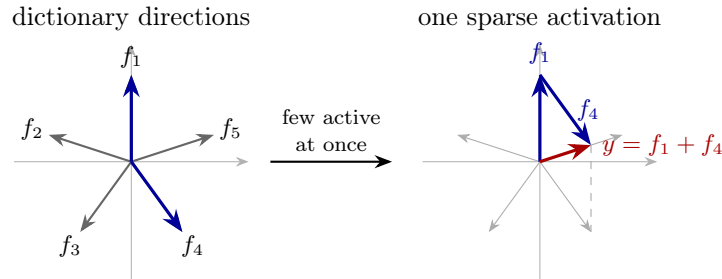


FIGURE 2. Superposition: five feature directions packed into a two-dimensional space as a regular pentagon. No two are orthogonal, but if only a few features are active at once, the activation is a sparse sum such as $y = f_1 + f_4$ and the active directions can still be recovered despite interference.

The nearest classical analogue is *compressed sensing* [10, 15]. Picture the features active on a given input as a sparse vector $x \in \mathbb{R}^m$ (only a few of the m possible features fire at once); the network stores it as $y = \Phi x$ in the $n \ll m$ dimensions of its *residual stream*—a linear *measurement*—and reading a feature back out is the *recovery* of x from y . Two classical theorems explain how this can work. First, there is room for the directions.

Theorem (Johnson–Lindenstrauss [24]). *For every $\varepsilon \in (0, 1)$ and every set of N points in $\mathbb{R}^d$, if $n \geq 8\varepsilon^{-2} \ln N$ then a suitably scaled random linear map $f: \mathbb{R}^d \rightarrow \mathbb{R}^n$ satisfies, with high probability and for all pairs at once,*

$$(1 - \varepsilon) \|x_i - x_j\|^2 \leq \|f(x_i) - f(x_j)\|^2 \leq (1 + \varepsilon) \|x_i - x_j\|^2.$$

In particular, $\mathbb{R}^n$ contains $N = e^{\Omega(\varepsilon^2 n)}$ unit vectors whose pairwise inner products are at most ε in absolute value: the number of ε -almost-orthogonal directions grows exponentially in the dimension.

And second, when only a few of those directions are active at once, the stored signal can be read back out. Say that $\Phi \in \mathbb{R}^{n \times m}$ is *restricted-isometric of order k* with distortion δ if

$$(1 - \delta)\|x\|^2 \leq \|\Phi x\|^2 \leq (1 + \delta)\|x\|^2$$

for every k -sparse vector x , meaning one with at most k nonzero entries.

Theorem (Candès [9]; sharp constant, Cai–Zhang [8]). *If Φ is restricted-isometric of order $2s$ with distortion $\delta < 1/\sqrt{2}$, then every s -sparse $x \in \mathbb{R}^m$ is the unique minimizer of*

$$\min_z \|z\|_1 \quad \text{subject to} \quad \Phi z = \Phi x,$$

and so is recovered exactly by a convex program. A random Φ (Gaussian or subgaussian entries) has the property with high probability once

$$n \gtrsim s \log(m/s),$$

That last scaling is the quantitative heart of superposition: the dimension a network needs grows with the number of features active *at once*, not with the total number it can represent. A few hundred dimensions can carry many thousands of features, provided only a handful fire on any given input. For trained [transformers](#) this remains an analogy rather than a theorem; turning it into one is part of the invitation.

There is a catch. Compressed sensing assumes the dictionary Φ is *known*. But in interpretability, someone hands you a trained network and asks what features it represents. You can run inputs through the model and record activation vectors, but the feature directions were not supplied with the weights. Recovering the features is therefore a problem in *dictionary learning*: from the observed activation vectors alone, simultaneously infer a dictionary Φ —a collection of feature directions (the columns of Φ , called *atoms*)—and, for each activation y , a sparse coefficient vector x with $y \approx \Phi x$. Both the dictionary and the codes are unknown; compressed sensing is the easier special case in which Φ is given and only x is sought.

A *sparse autoencoder* does this empirically: trained to reconstruct activations under an ℓ^1 penalty, it learns an overcomplete dictionary whose features are often more interpretable than the network’s raw neurons [6, 14]. That such structure *emerges* from a bare reconstruction-plus-sparsity objective is an empirical observation, not a theorem. The theorem we would like is an identifiability theorem: when does this procedure recover the network’s own features rather than a prettier coordinate system imposed by the researcher?

A second hypothesis says that many concepts are literally directions and that composition is addition. Park, Choe, and Veitch [38] make this *linear representation hypothesis* precise for next-[token](#) prediction: they distinguish directions used to *read* a concept from directions used to *steer* it, and show that the two can be identified through a non-Euclidean geometry. If there is a *causal inner product* in

which independently varying concepts are orthogonal, then the associated Riesz map sends readout directions to steering directions. They give an estimator for this geometry from the unembedding matrix and test it on clean word-pair concepts in LLaMA-2. For arbitrary internal features, the existence and identifiability of such a geometry remain open.

Arditi et al. [2] show that in several open chat models, refusal is mediated by a single residual-stream direction: this direction both predicts whether the model will refuse and causally controls refusal when added or removed. A lone direction serving as both readout and control is evidence for the linear representation hypothesis, though in a different representation space from the unembedding geometry above.

Representation theory appears in trained arithmetic circuits. A one-layer transformer trained to add integers modulo p learns a discrete-Fourier “clock” algorithm, with the characters of $\mathbb{Z}/p\mathbb{Z}$ appearing in its [weights](#) [34]. Chughtai, Chan, and Nanda [11] take the first step beyond this abelian case. The ambient structure is the Artin–Wedderburn decomposition of the group algebra into matrix blocks,

$$\mathbb{C}[G] \cong \bigoplus_{\rho \in \widehat{G}} M_{d_\rho}(\mathbb{C}), \quad \sum_{\rho \in \widehat{G}} d_\rho^2 = |G|,$$

one block per irreducible representation ρ . For a nonabelian group the higher-dimensional blocks are genuine matrix representations, and trained networks compute the product through a sparse subset of them. The algorithm’s correctness is a representation-theory theorem; that networks implement it is empirical. Which irreducible representations training selects is still something we discover after the fact.

The safety question is algorithmic selection. A trained network can discover a crisp algebraic algorithm, hide it in weights, and use only part of the mathematically available structure. If we want to predict a system’s behavior off-distribution, it matters which algorithm training found, not only that the training [loss](#) is low.

One caveat carries across this whole enterprise: a direction that *predicts* a feature is only a *correlate*. Showing that the network actually *uses* it takes an intervention—ablating the direction, or patching it to the value it would take on another input, and watching the behavior change—not a [probe](#) that merely reads it off.

4.2. Open problems.

★ **Open Problem 3** (The geometry and identifiability of superposition). Classical dictionary learning already has identifiability theorems; the open problem is to prove the version mechanistic interpretability actually needs. Suppose activations have the form $y = \Phi x + \eta$, where the columns of Φ are feature directions, the sparse codes x have correlated supports, and the recovery method is an idealized sparse-autoencoder objective rather than an oracle sparse-coding algorithm. Characterize, in terms of coherence $\mu = \max_{i \neq j} |\langle v_i, v_j \rangle|$, sparsity, sample size, and support correlations, when the learned dictionary recovers the original directions up to permutation and scaling, and when it instead *merges* co-occurring directions or *splits* one direction into several. The clean first case is a random incoherent dictionary, exactly sparse nonnegative coefficients, and no optimization error.

★ **Open Problem 4** (Training for interpretability). Post-hoc interpretability asks whether a trained model’s features can be recovered after the fact. A complementary problem is to train models whose features are easier to recover in the first place. Find objectives, architectures, or regularizers that induce internal representations which are sparse, modular, and stable enough to interpret, without destroying performance on the task. A first mathematical target is a toy model in which one can prove that adding an interpretability regularizer selects a dictionary with lower coherence or more identifiable atoms than ordinary empirical-risk minimization.

★ **Open Problem 5** (Representation theory of learned circuits). Networks trained on modular addition often learn Fourier or representation-theoretic algorithms, but the precise circuit can depend on architecture, initialization, and training dynamics. Fix a finite group G , a data distribution on pairs (a, b) , and a small architecture. Prove either a selection theorem—which irreducible representations are learned, with what multiplicities, under a specified initialization and optimizer—or a non-identifiability theorem showing that several inequivalent representation-theoretic circuits are equally stable minima. The first foothold is $G = S_3$ with the architecture fixed in advance.

A broad, structured catalogue of open problems in this area—several with a clean mathematical core—is collected by Sharkey et al. [46].

5. ANALYSIS AND GEOMETRY: WHICH SOLUTION DOES TRAINING FIND?

5.1. Why trained networks generalize: singular learning theory. A trained network is a setting of parameters chosen to fit its *training data*—the finite set of examples it was optimized on. The trouble is that a large network has astronomically many settings that fit those examples equally perfectly, and they disagree wildly everywhere else; the optimizer—*gradient descent*, which repeatedly nudges the parameters downhill on the training error—must land on one of them. For safety, the selected solution matters: a model that behaved well while we were watching may behave differently once deployed. So why do the solutions found by training *generalize*—predict well on fresh data from the same source—when the overwhelming majority of data-fitting solutions would not? These networks reach zero error even when the labels are replaced by pure noise, and a two-layer network with only $2n + d$ parameters can fit *any* labeling of n points in $\mathbb{R}^d$ (Zhang et al. [57]); the classical bounds that explain generalization by counting parameters say nothing here.

A different and more general answer comes from *singular learning theory* [54]. Realistic models are *singular*: distinct parameters compute the same function, so the optimal set is not an isolated point but a positive-dimensional—and typically singular—variety, flat to second order along its own directions. The classical large-sample asymptotics all assume an isolated, curved optimum; the right replacements come from algebraic geometry. One caveat: the theory is exact for *Bayesian* learning, not directly for stochastic gradient descent (gradient descent driven by

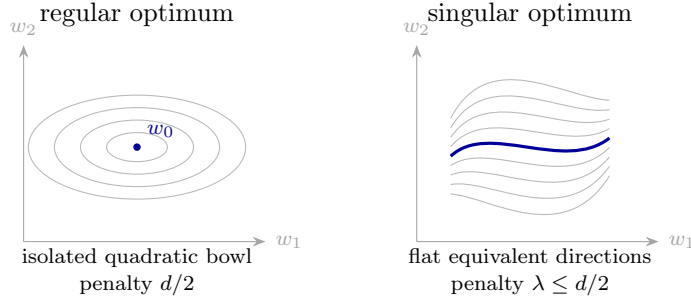


FIGURE 3. Regular asymptotics treat the optimum as an isolated quadratic bowl, giving the classical $\frac{d}{2} \log n$ complexity penalty. Singular learning theory allows a whole variety of parameters to compute the same function; the effective penalty is λ , often strictly smaller than $d/2$.

noisy, small-batch estimates of the gradient)—bridging the two is the open problem that closes this section.

Make it precise. Fix a model $p(x | w)$ with parameter w in a compact $W \subseteq \mathbb{R}^d$, a prior φ on W (a probability density encoding which parameters are plausible before any data is seen), and a true distribution $q(x) = p(x | w_0)$ lying in the model. Write

$$K(w) = \text{KL}(q \| p(\cdot | w)) = \int q(x) \log \frac{q(x)}{p(x | w)} dx,$$

so that the set of *true parameters* $W_0 = K^{-1}(0)$ is exactly where the model recovers q . Given n i.i.d. samples $X_1, \dots, X_n$, the central object is the *Bayes free energy*—equivalently the negative log marginal likelihood, the codelength the model assigns to the data,

$$\bar{F}_n = -\log \int_W \prod_{i=1}^n p(X_i | w) \varphi(w) dw.$$

(The bar is part of the name; $\bar{F}_n$ is the random free energy of the sample, not an expectation.) Normalizing the same integrand gives the *posterior* $\propto \prod_i p(X_i | w) \varphi(w)$ —the prior reweighted by how well each parameter fits the data—so $\bar{F}_n$ is minus the log of its normalizer. Take first the *regular* case: the model is *identifiable* (distinct parameters give distinct distributions) and the optimal set is a single point $W_0 = \{w_0\}$ at which the Hessian of K —the *Fisher information* $I(w_0)$, the matrix of second derivatives of the divergence K at the optimum—is positive-definite. Then the integrand defining $\bar{F}_n$ concentrates in a shrinking neighborhood of w_0 ; expanding $\log p(X_i | w)$ to second order there makes the integral Gaussian, contributing a volume factor $(2\pi/n)^{d/2} \det I(w_0)^{-1/2}$. Taking $-\log$,

$$\bar{F}_n = nS_n + \frac{d}{2} \log n + O_p(1), \quad S_n = -\frac{1}{n} \sum_i \log q(X_i),$$

where S_n is the empirical entropy and the remainder is bounded in probability.² The complexity penalty is exactly half the parameter count. Watanabe’s theorem is the singular replacement, in which $d/2$ gives way to a birational invariant.

Theorem (Watanabe’s free-energy asymptotics [54]). *For a singular model satisfying Watanabe’s regularity conditions (real-analyticity and a mild variance bound³), as $n \rightarrow \infty$,*

$$\bar{F}_n = n S_n + \lambda \log n - (m - 1) \log \log n + O_p(1),$$

where the learning coefficient $\lambda > 0$ —the real log canonical threshold (RLCT) of K relative to the prior φ —and the integer $m \in \{1, \dots, d\}$, its multiplicity, take the places of the $d/2$ and the 1 of the regular case. The $O_p(1)$ remainder converges in distribution.⁴ The same coefficient controls generalization: the expected generalization error G_n —the extra average log-loss the posterior-averaged model incurs on a fresh sample—obeys $\mathbb{E}[G_n] = \lambda/n + o(1/n)$.

The learning coefficient has two faces, one analytic and one geometric. Analytically, $-\lambda$ is the largest pole of the zeta function

$$\zeta(z) = \int_W K(w)^z \varphi(w) dw,$$

holomorphic for $\Re z > 0$ and meromorphically continued to $\mathbb{C}$ with poles on the negative rational axis; $-\lambda$ is the pole nearest the origin and m its order. Geometrically, λ is computed by Hironaka’s *resolution of singularities* [21]: a change of coordinates turns K into a monomial, and λ is read off from its exponents as a minimum of rational ratios—the bridge from algebraic geometry to the learning rate. In the regular case the resolution is trivial, $\lambda = d/2$ and $m = 1$, and the theorem recovers the regular asymptotic expansion above; in a singular model $\lambda \leq d/2$, often strictly. Because the same λ sets the generalization error λ/n , smaller λ means better generalization, and the singular points of smallest λ are exactly the ones whose share of the posterior grows with n . The Bayesian posterior therefore concentrates, on its own, on the best-generalizing solutions. This is the precise content of the slogan that learning prefers “simpler” solutions: simplicity here means a more degenerate geometry, not fewer parameters. That degeneracy is largely *structural*: it comes from directions in parameter space along which the function does not change—for instance those freed when an internal unit is effectively switched off—and these are exactly what pull λ below $d/2$. The symmetries are not a defect to be engineered

² $O_p(1)$ means *bounded in probability*: for every $\varepsilon > 0$ there is an M with $\Pr(|\cdot| > M) < \varepsilon$ for all large n .

³The *relatively finite variance* condition: writing $f(x, w) = \log(q(x)/p(x | w))$, $K(w) = \mathbb{E}_q f(X, w)$, and $V(w) = \mathbb{E}_q f(X, w)^2$, one assumes $V(w) \leq C K(w)$ near the optimal set—the variance of the log-likelihood ratio is controlled by its mean.

⁴In the regular case the limit is $C - \frac{1}{2}\chi_d^2$ for a constant C —the Bayesian shadow of Wilks’ theorem, since the log-likelihood-ratio statistic $2(\sup_w \sum_i \log p(X_i | w) - \sum_i \log p(X_i | w_0)) \Rightarrow \chi_d^2$. In a singular model the limit is in general non-Gaussian, a functional of a Gaussian process on the resolution manifold U described below.

away. A model forced to be regular, every degeneracy broken so that $\lambda = d/2$, would generalize at the worse classical rate $d/(2n)$.

The coefficient λ need not be found by hand. Introduce a knob $\beta > 0$, an *inverse temperature*, and the *tempered posterior*

$$p_\beta(w) \propto \varphi(w) e^{-\beta n L_n(w)}, \quad L_n(w) = -\frac{1}{n} \sum_i \log p(X_i | w),$$

which interpolates between the prior ($\beta = 0$) and concentration on the best-fitting parameters ($\beta \rightarrow \infty$); write $Z(\beta)$ for its normalizing constant and $F(\beta) = -\log Z(\beta)$ for the associated free energy. The learning coefficient is then a derivative of F : it measures how the posterior-averaged loss responds as β varies, and so can be *estimated* by sampling the tempered posterior, needing neither the true distribution nor an explicit resolution of singularities [55]. That practicality is what lets the theory meet networks far too large to analyze by hand.

What one estimates is not the global λ but the *local learning coefficient*: the same invariant computed in a neighborhood of the particular solution w^* that training actually reached, measuring how degenerate *that* solution is [27]. As a network trains, its local learning coefficient jumps at discrete moments—*phase transitions*, where the solution moves to a qualitatively new, differently-degenerate part of the landscape as a new piece of internal structure forms. Tracking these jumps is the program of *developmental interpretability*: so far on relatively small models and with preliminary validation, though the same estimators have already surfaced tens of thousands of candidate structures inside a billion-parameter language model [33]. Two networks can fit the training data identically and yet compute by different internal mechanisms, hence behave differently on inputs outside that data—precisely what ordinary testing, which only probes behavior on data, cannot detect. The singular geometry governs *which* mechanism training selects; if it favors mechanisms that are cheap to fit but hard to read, or that break under a shift in the input distribution, then λ helps decide whether aligned-looking behavior is stable and legible or an artifact about to fail [28].

★ **Open Problem 6** (Learning coefficients of real architectures). Compute or bound the RLCT λ (and its multiplicity m) for singularities of explicitly specified architectures not already covered by classical singular models—say a one-head [attention](#)-only transformer on a finite vocabulary, or a small network trained on modular addition. Relate the answer to interpretable structure in the learned solution, and compare it to the empirically estimated local learning coefficient. Exact RLCTs are known for several classical singular statistical models; even tight bounds for these neural examples would sharpen both the generalization theory and the developmental picture.

Training does not always lower the loss smoothly; long flat stretches give way to sudden drops. The most striking example is *grokking*: a network trained on modular arithmetic (the algebra of Section 4) fits its training data early but generalizes only much later, and abruptly.⁵ The cleanest solvable model of such stepwise learning

⁵Grokking and the staircase below are cousins, not the same phenomenon: grokking is a delayed jump in performance on *held-out* data after the training loss is already low, whereas the staircase

is the deep *linear* network: a product of matrix factors $W_L \cdots W_1$ fit to a target linear map by *gradient flow* (gradient descent in continuous time) on squared error. The composite map is linear, but as a function of the separate factors the loss is a non-convex polynomial, so the flow is genuinely nonlinear. Saxe, McClelland, and Ganguli [44] show that when the inputs are *whitened* (linearly transformed to have identity covariance), the singular value decomposition of the input–output correlation matrix *decouples* the dynamics into independent scalar equations, one per singular value s : each is the logistic ordinary differential equation $\tau \dot{u} = 2u(s - u)$, whose solution rises sigmoidally from near 0 to s in time of order τ/s (Figure 4). Two consequences are exact: the modes are learned *one at a time, largest s first*; and the training error falls not smoothly but in a staircase—long plateaus near saddle points of the loss, broken by sharp drops as each mode is acquired. Here “which structure the network learns, and in what order” has a closed-form answer. The staircase survives beyond the linear case: Abbe, Boix-Adserà, and Misiakiewicz [1] prove that training a two-layer network on a *sparse* target—one depending on only a few of its inputs—again proceeds in stages, saddle to saddle, with the waiting time before each stage set by a combinatorial *leap complexity* measuring how many inputs must be assembled at once. Carrying such exact accounts to the nonlinear networks used in practice is open.

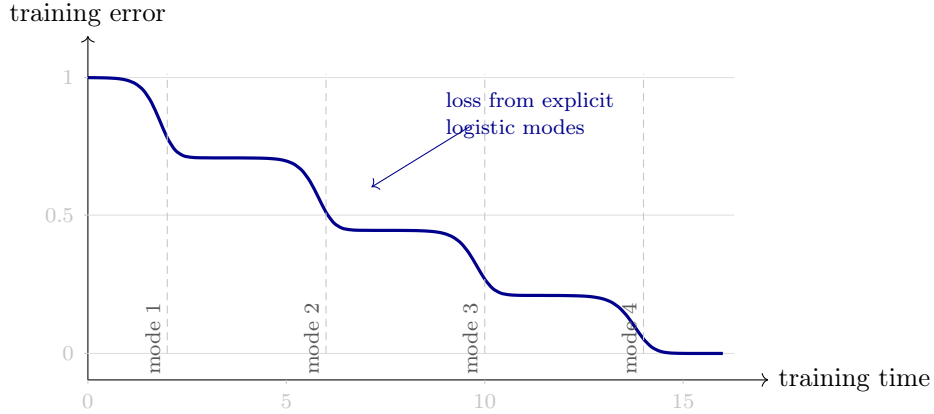


FIGURE 4. A learning staircase computed from the deep-linear solution of Saxe et al. In mode i , s_i is the target singular value and $u_i(t)$ is the coefficient learned by time t , satisfying $\tau \dot{u}_i = 2u_i(s_i - u_i)$. The curve plots normalized training error $\sum_i (s_i - u_i(t))^2$ for $\tau = 0.5$ and $s = (1, 0.95, 0.9, 0.85)$, with small initial coefficients chosen so the four modes turn on at separated times. Each mode turning on produces one drop.

★ **Open Problem 7** (Effective loss and learning dynamics). Watanabe’s theorem is Bayesian; gradient descent is not. For stochastic gradient descent or gradient

is a stepwise fall of the *training* loss itself. Both are read, in this program, as transitions between differently-singular regions of parameter space.

flow near a singular optimal set stratified by learning coefficient, derive an effective dynamics on the strata: which stratum is selected, on what timescale, and how noise or finite learning rate changes the selection. The solvable deep-linear staircase above is the calibration case: any proposed theory should recover its stage order and timescales, then predict when a nonlinear network moves from one computational mechanism to another.

5.2. Generalization beyond the training distribution. The theory above concerns generalization *in distribution*: test data drawn from the same law as the training data. The safety-critical regime is the opposite one, *out of distribution*—how a trained system behaves on inputs unlike anything it saw in training—and there the gap between fitting and understanding reopens. Two parameter settings with identical training loss can implement different mechanisms (as above) and therefore extrapolate differently; which extrapolation gradient descent selects is, at present, something we observe after the fact rather than predict. In *goal misgeneralization* [26], a system under *distribution shift*—a deployment input distribution unlike the one it trained on—keeps its *capabilities* while pursuing a *different goal* from the one intended—the coin-collector trained where the coin always sat at the level’s right end, which learned “move right” rather than “reach the coin.” Capable but misdirected is the dangerous quadrant, and it is invisible to any testing that stays in distribution.

★ **Open Problem 8** (A predictive theory of out-of-distribution generalization). Given a training distribution, an architecture, and a training procedure, *predict*—rather than measure after the fact—which loss-equivalent extrapolation training selects off-distribution. Start with a toy goal-misgeneralization environment having two perfect training policies (rules assigning an action to each situation), one tracking the intended goal and one tracking a proxy that agrees on the training distribution. Prove which policy gradient descent selects as a function of initialization, architecture, and data distribution; then ask when a distribution shift preserves capabilities while breaking the intended goal.

6. GETTING STARTED IN AI SAFETY

Technological progress is a human choice, not a law of nature. When mathematical ideas help build systems whose effects will be felt at civilizational scale, truth-seeking remains a core value, but it is no longer the only value in view. The civic mathematician also asks what our truths are for: whether they can help design AI systems to be more legible, more accountable, and easier to steer toward coexistence with us.

The open problems above are meant as entry points into the new field of Math for AI Safety. I hope you’ll pick one whose language already feels like home, strip it down to the simplest interesting case, and start proving things! Other ways in are workshops like [ILIAD](#), research programs like [MATS](#) and [SPAR](#), and organizations like [CHAI](#), [Constellation](#), [FAR AI](#), and [Sequent](#).

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APPENDIX A. A GLOSSARY FOR MATHEMATICIANS

<i>token</i>	the atomic unit (word or word-fragment) a language model reads and predicts.
<i>loss</i>	the scalar a network is trained to minimize; for language models, the negative log-probability it assigned to the actual next token.
<i>weights</i>	the trainable parameters of an artificial neural network: the entries of its matrices, trained by gradient descent and fixed at deployment time.
<i>activations</i>	the values that flow through the network on a given input (the outputs of its neurons and layers), as opposed to the fixed weights.
<i>gradient descent</i>	the optimizer: repeatedly nudge the parameters θ against the gradient of the loss, $\theta \leftarrow \theta - \eta \nabla L(\theta)$.
<i>ReLU</i>	the rectified linear unit $x \mapsto \max(0, x)$, a canonical piecewise-linear nonlinearity; modern transformers often use smooth or gated variants instead (GELU, SiLU, SwiGLU).
<i>transformer</i>	the dominant network architecture for language: a stack of layers that mix information across token positions by <i>attention</i> and read from / write to the residual stream.
<i>attention (head)</i>	the transformer operation by which each position selectively reads from the others; one such unit is an attention <i>head</i> .
<i>residual stream</i>	the running vector of activations a transformer reads from and writes to at each layer; the space in which “features” live.
<i>feature</i>	a hypothesized variable or property a network represents in its activations (e.g. “the text is in French”); it may be distributed, nonlinear, or not human-interpretable—identifying features with human-meaningful directions is itself a hypothesis, not a definition.
<i>probe</i>	a simple (usually linear) function of a network’s activations, trained or searched to read off some quantity of interest.
<i>policy</i>	in reinforcement learning, an agent’s rule for choosing actions from states (a possibly randomized map from states to actions).

<i>superposition</i>	a hypothesized representation scheme in which a network stores more features than it has dimensions, as non-orthogonal directions, tolerable when few features are active at once.
<i>interpretability</i>	reverse-engineering a trained network into human-understandable structure (circuits, features, algorithms), validated by intervention and not by human-readable description alone.

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